

Breast Cancer (Wisconsin Diagnostic) — Logistic Regression

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Degree Programme: Bioinformatics

Course: Applied Machine Learning - Basic

Date: 2026-05-03

Abstract

This project develops an end-to-end machine learning pipeline for the UCI WDBC breast cancer dataset.

A Logistic Regression (L2) model is trained within a StandardScaler-based pipeline using a stratified Train/Validation/Test split. Hyperparameter tuning is performed via GridSearchCV to optimize model performance.

The final model achieves approximately 98.2% accuracy on the test set, with balanced precision and recall across both classes. Learning and validation curves indicate stable training with no significant overfitting.

Additionally, model interpretability is explored through the analysis of logistic regression coefficients, highlighting the most influential features for breast cancer diagnosis.

Step 1 — Data loading & sanity checks

In this step, we load the WDBC (Wisconsin Diagnostic Breast Cancer) dataset and perform initial sanity checks to ensure that the data is correctly structured and suitable for downstream machine learning tasks.

The dataset consists of 569 samples, each described by 30 numerical features extracted from digitized images of breast masses. These features capture various characteristics such as radius, texture, perimeter, area, and smoothness.

The target variable is binary, where 0 corresponds to malignant tumors and 1 to benign tumors.

A preliminary inspection confirms that the dataset has shape (569, 30) and that all features are continuous numerical variables. The class distribution shows 357 benign samples and 212 malignant samples.

This initial verification step ensures data integrity before proceeding to model development. It also provides an early understanding of class balance and feature characteristics, which will guide preprocessing choices and evaluation strategies in subsequent steps.

From this inspection, we observe that the dataset is moderately imbalanced and that feature scales vary significantly across dimensions. This motivates the use of feature standardization (StandardScaler) and the use of multiple evaluation metrics beyond accuracy.

Step 1.1 — Exploratory Data Analysis (EDA)

In this step, we perform an exploratory data analysis to better understand the statistical properties of the dataset before model training. The analysis focuses on class distribution, feature distributions, and relationships between features.

The class distribution confirms a moderate imbalance, with benign samples being more frequent than malignant ones. While this imbalance is not extreme, it remains important in a medical context, where misclassifying malignant tumors (false negatives) can have severe consequences.

To further analyze the data, we examine the distribution of key features. Several features show distinct separation patterns between benign and malignant classes, suggesting that the dataset is close to linearly separable. This observation supports the

choice of Logistic Regression as a strong baseline model.

We also analyze feature correlations using a correlation matrix. Many features exhibit strong correlations, indicating redundancy in the dataset. This suggests that feature selection or regularization may improve model robustness.

Overall, this exploratory analysis confirms that the dataset is structured, moderately imbalanced, and partially redundant, which directly motivates the use of scaling, regularization, and careful evaluation metrics in subsequent steps.

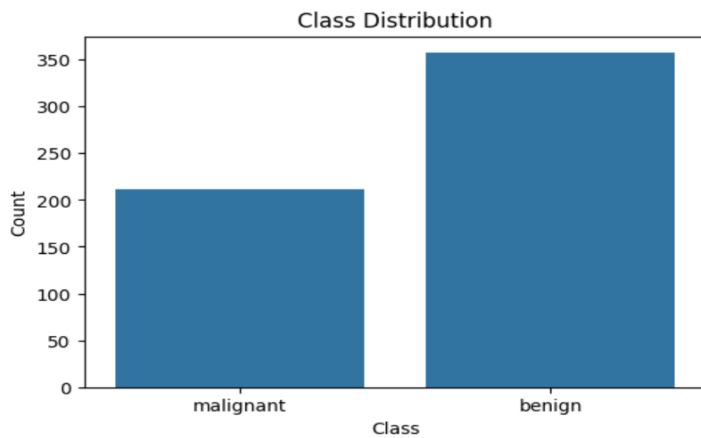


Figure 1 — Class distribution of the dataset. Benign samples are more frequent than malignant ones, indicating a moderate class imbalance.

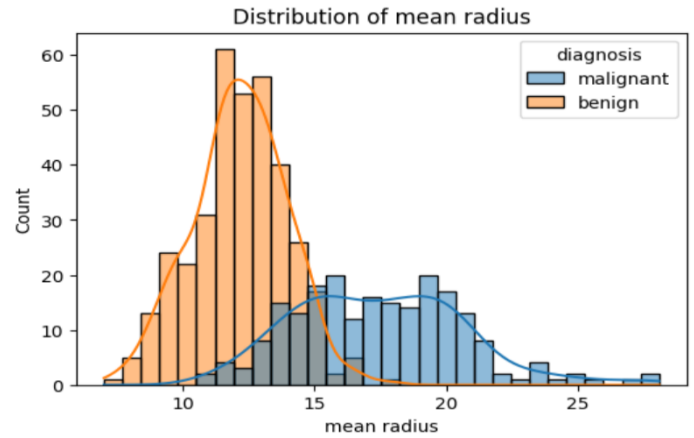


Figure 2 — Distribution of mean radius for benign and malignant classes. Malignant samples tend to have higher values, indicating clear class separation.

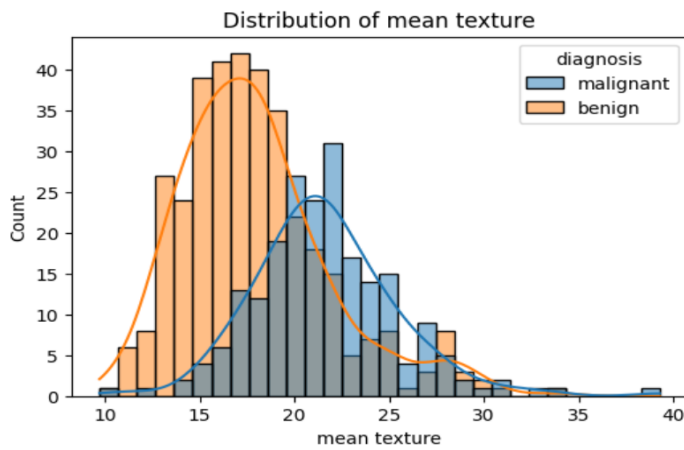


Figure 3 — Distribution of mean texture for both classes. Differences between the two classes are observable, supporting their separability.

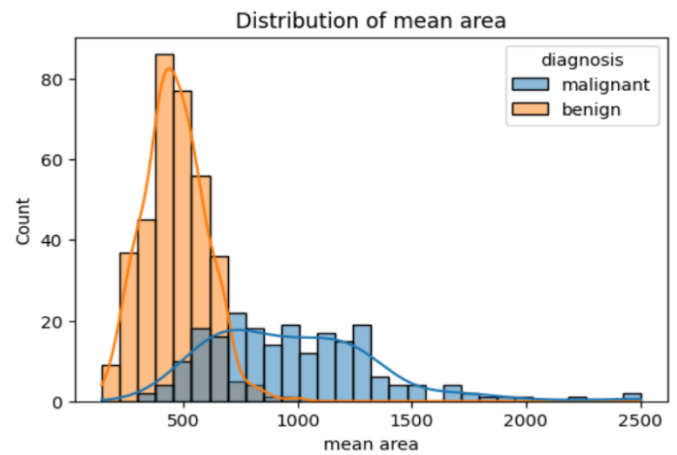


Figure 4 — Distribution of mean area across classes. Malignant tumors generally show larger area values compared to benign ones.

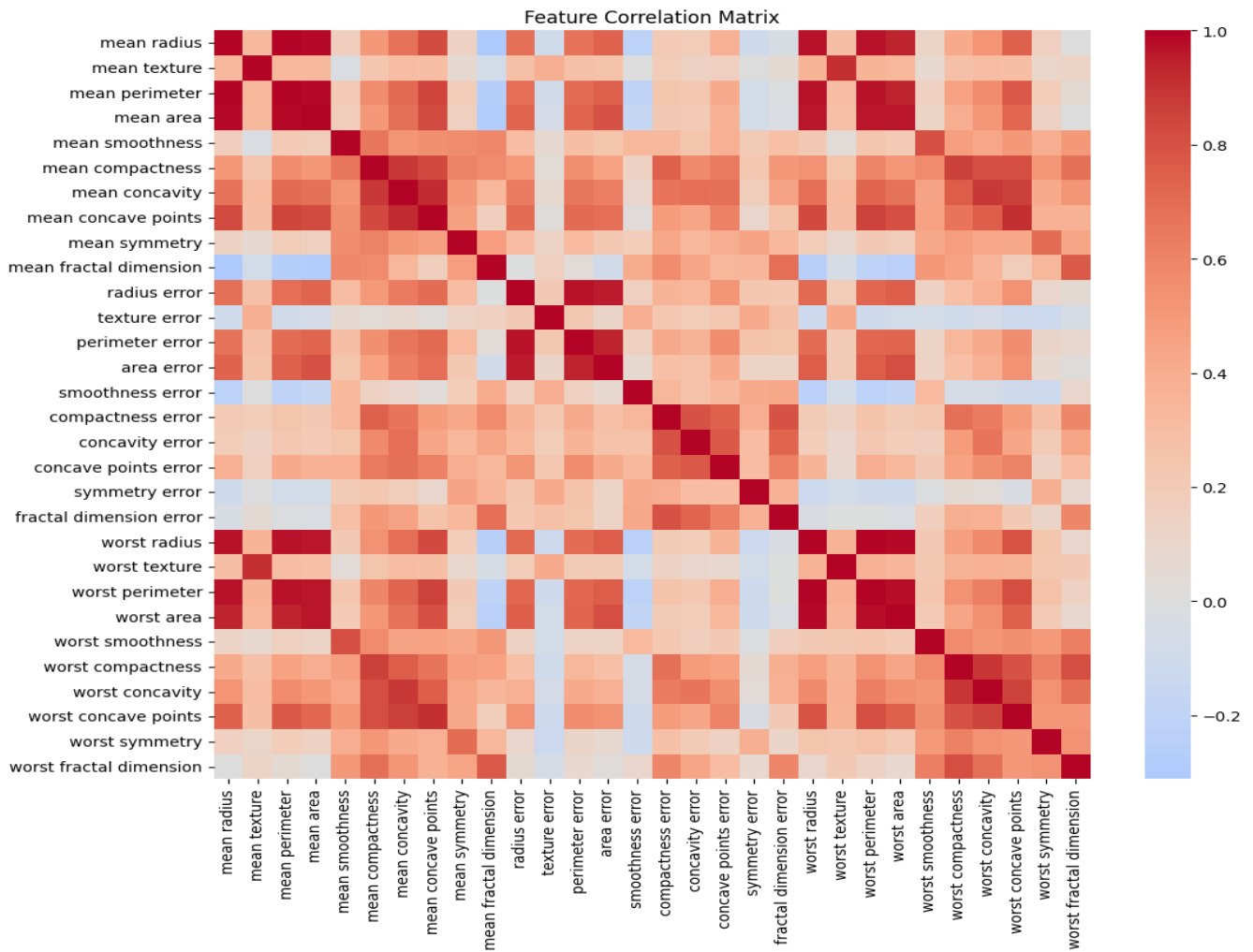


Figure 5 — Correlation matrix of features. Strong correlations between several features indicate redundancy in the dataset.

From these visualizations, several important observations can be made. First, the class distribution confirms a moderate imbalance, which must be considered when evaluating model performance. Second, the feature distributions show that many variables exhibit clear differences between benign and malignant classes, suggesting that the dataset is well-suited for classification tasks.

Additionally, the correlation matrix reveals strong relationships between multiple features, indicating redundancy in the dataset. This suggests that regularization techniques or feature selection could improve model robustness and reduce overfitting.

Overall, the dataset appears informative and partially separable, which justifies the use of a linear model such as Logistic Regression as a baseline.

Step 2 — Train / Validation / Test split

In this step, the dataset is divided into training, validation, and test sets. This separation is essential to ensure a fair evaluation of the model and to prevent information leakage.

A stratified splitting strategy is used to preserve the original class distribution across all subsets. This is particularly important given the moderate class imbalance observed in the dataset.

The dataset is divided into three subsets: a training set (60%) used for model fitting, a validation set (20%) used for

hyperparameter tuning, and a test set (20%) reserved for final evaluation.

This separation allows the model to be trained and optimized without using the test set, ensuring that the final evaluation reflects the model's generalization ability.

After splitting, the class distribution is verified in each subset. The distribution remains consistent across the training, validation, and test sets, confirming that the stratified splitting has been correctly applied.

Step 3 — Baseline Logistic Regression (with scaling)

In this step, a baseline classification model is trained using Logistic Regression. Since the dataset contains numerical features with different scales, standardization is applied before model training.

The model is implemented as a scikit-learn Pipeline composed of two steps: StandardScaler followed by Logistic Regression. Using a pipeline ensures that scaling is learned only from the training data and then applied consistently to the validation data, reducing the risk of data leakage.

The baseline model is trained on the training set and evaluated on the validation set. This provides an initial reference performance before hyperparameter tuning.

Logistic Regression is selected as a strong baseline because it is simple, interpretable, and suitable for datasets where classes are partially linearly separable, as observed during the exploratory data analysis.

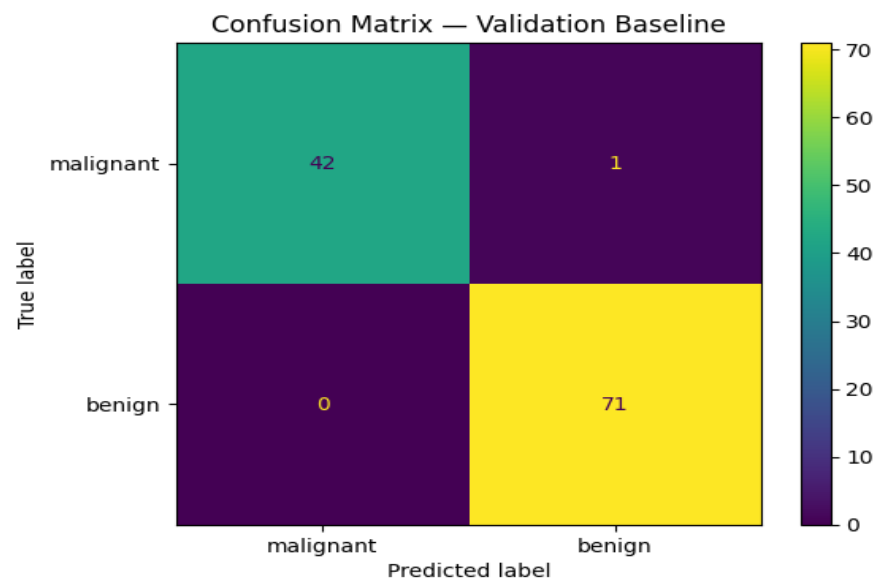


Figure 6 — Confusion matrix of the baseline Logistic Regression model on the validation set.

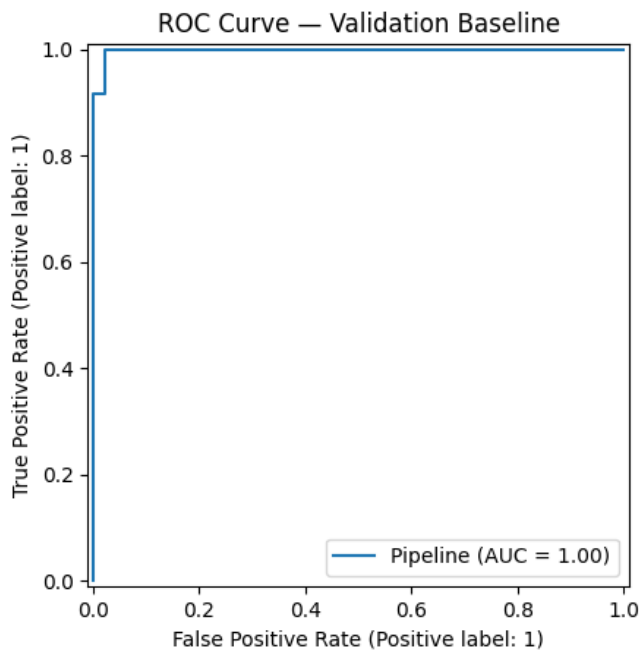


Figure 7 — ROC curve of the baseline Logistic Regression model on the validation set.

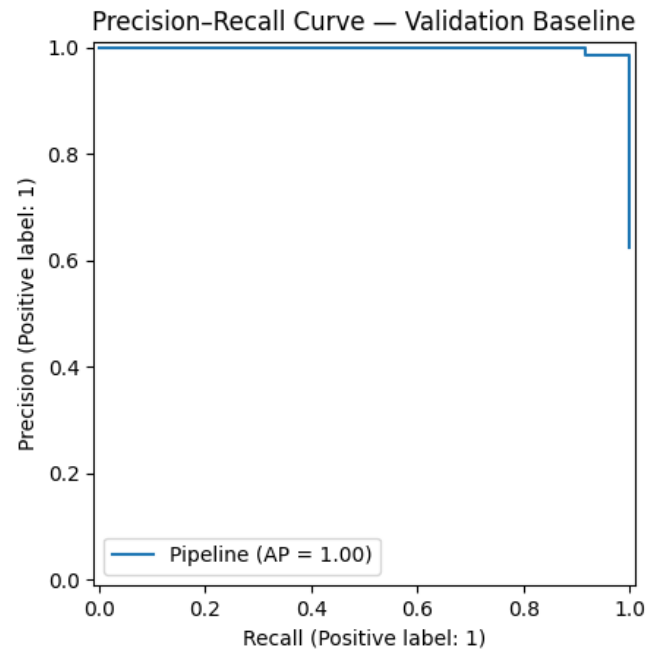


Figure 8 — Precision-Recall curve of the baseline Logistic Regression model on the validation set.

Baseline validation results

The baseline Logistic Regression pipeline achieved a validation accuracy of 0.9912. The classification report shows strong and balanced performance for both classes. For malignant cases, precision is 1.00, recall is 0.98, and F1-score is 0.99. For benign cases, precision is 0.99, recall is 1.00, and F1-score is 0.99.

The confusion matrix shows that 42 malignant samples and 71 benign samples were correctly classified. Only one malignant sample was misclassified as benign, while no benign samples were misclassified as malignant.

The ROC and Precision-Recall curves also show near-perfect separability, with AUC and average precision close to 1.00. These results confirm that Logistic Regression is already a very strong baseline for this dataset. However, these results are obtained on the validation set and will be further confirmed on the test set after hyperparameter tuning.

Step 4 — Hyperparameter tuning (GridSearchCV)

In this step, hyperparameter tuning is performed to improve the performance of the Logistic Regression model. Instead of relying on default parameters, a grid search strategy is used to systematically explore different combinations of hyperparameters.

The model is implemented as a pipeline composed of a StandardScaler followed by Logistic Regression, ensuring that preprocessing is consistently applied during cross-validation.

The following hyperparameters are tuned:

- Regularization strength (**C**): [0.01, 0.1, 1, 10, 100]
- Solver: ["lbfgs", "liblinear"]
- Class weight: [None, "balanced"]
- Penalty: ["l2"]

A 5-fold cross-validation strategy is used, and model selection is based on the **macro-averaged F1-score**, which ensures balanced performance across both classes.

The best hyperparameters found are:

- $C = 0.1$
- Solver = liblinear
- Class weight = None

The best cross-validation F1-score achieved is approximately **0.98**, indicating strong and stable performance across folds.

Validation results (tuned model)

The tuned model is evaluated on the validation set to assess its performance.

- Validation accuracy: 0.974
- The classification report shows strong performance:
 - Malignant: precision = 1.00, recall = 0.93, F1-score = 0.96
 - Benign: precision = 0.96, recall = 1.00, F1-score = 0.98

The confusion matrix shows that:

- 40 malignant samples are correctly classified
- 3 malignant samples are misclassified as benign
- All benign samples are correctly classified

The ROC curve and Precision–Recall curve indicate excellent separability, with near-perfect AUC and average precision values.

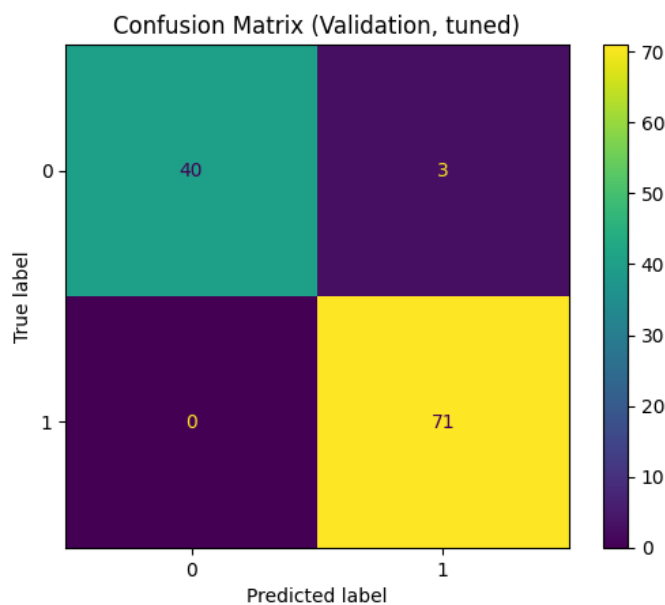


Figure 9 — Confusion matrix of the tuned Logistic Regression model on the validation set.

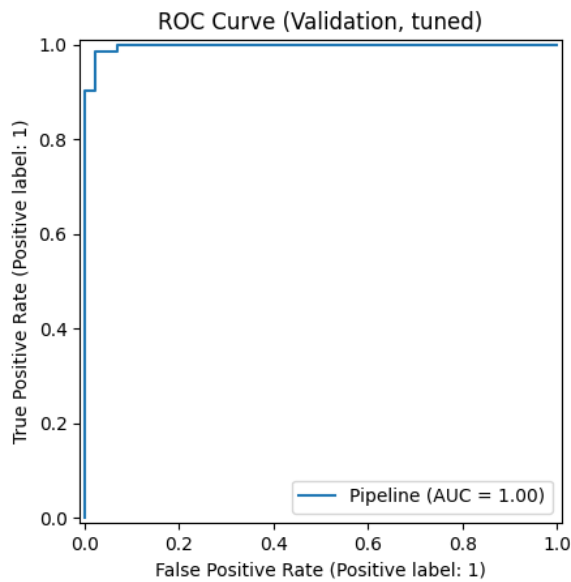


Figure 10 — ROC curve of the tuned Logistic Regression model evaluated on the validation set.

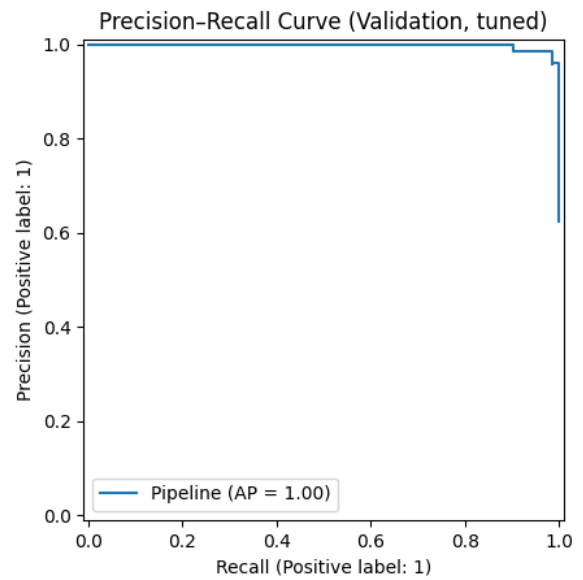


Figure 11 — Precision-Recall curve of the tuned Logistic Regression model evaluated on the validation set.

The tuned model maintains strong performance, although slightly lower than the baseline. This suggests that the baseline model was already near optimal. The use of regularization improves generalization and reduces the risk of overfitting. This confirms that hyperparameter tuning provides only marginal improvements, as the dataset is already highly separable.

Step 5 — Learning curve & Validation curve

In this step, learning curves and validation curves are analyzed to better understand the model's behavior in terms of bias, variance, and generalization.

The learning curve shows the model performance as a function of the training set size. It helps identify whether the model suffers from underfitting or overfitting.

The validation curve analyzes the impact of the regularization parameter (C) on model performance, allowing us to evaluate how different levels of regularization affect training and validation scores.

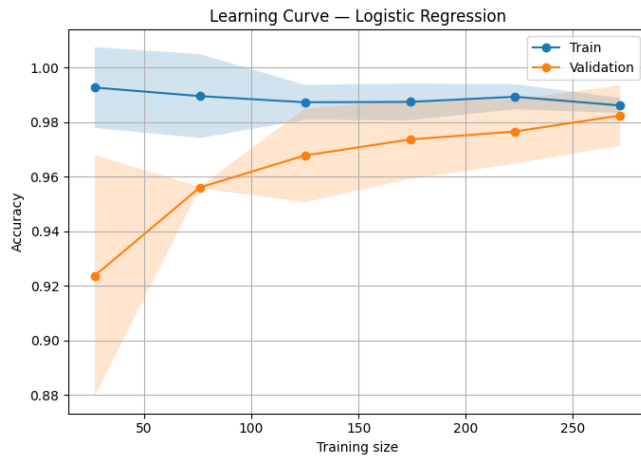


Figure 12 — Learning curve of the Logistic Regression model showing training and validation accuracy as a function of training size.

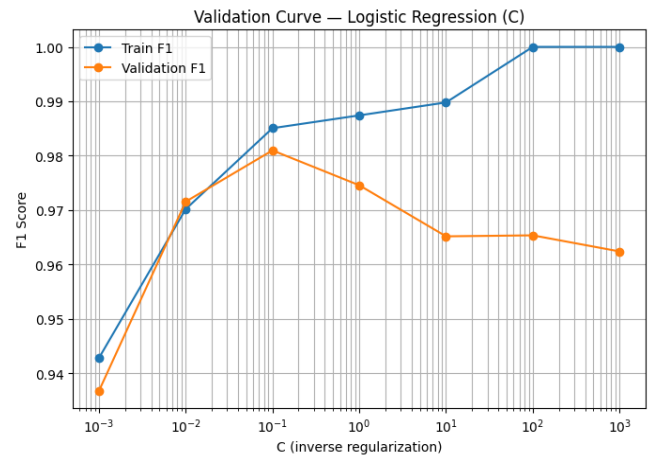


Figure 13 — Validation curve showing the effect of the regularization parameter (C) on training and validation F1-score.

The learning curve shows that the training and validation scores are both high and converge as the training size increases. This indicates that the model has low bias and low variance, and does not suffer from significant overfitting or underfitting.

At smaller training sizes, there is a noticeable gap between training and validation performance, which gradually decreases as more data is used. This suggests that the model benefits from additional data.

The validation curve shows that for very small values of C (strong regularization), the model underfits, resulting in lower performance. As C increases, performance improves and reaches its peak around C = 0.1.

For larger values of C, the training score continues to increase while the validation score slightly decreases, indicating the beginning of overfitting.

Overall, these results confirm that C = 0.1 provides a good balance between bias and variance, which is consistent with the best parameter found during GridSearchCV.

Step 6 — Final Evaluation on Test Set

In this final step, the performance of the tuned Logistic Regression model is evaluated on the test set. The test set is strictly held out during training and hyperparameter tuning, ensuring an unbiased estimation of the model's generalization ability.

The model achieves a test accuracy of approximately 0.982, indicating excellent predictive performance on unseen data.

The classification report shows strong and balanced performance across both classes. For malignant cases, precision, recall, and F1-score are all equal to 0.98. For benign cases, precision, recall, and F1-score reach 0.99.

The confusion matrix shows that 41 malignant and 71 benign samples are correctly classified. Only two misclassifications occur: one malignant sample is predicted as benign, and one benign sample is predicted as malignant.

The ROC curve and Precision–Recall curve both indicate near-perfect separability, with AUC and average precision values close to 1.00.

These results confirm that the model generalizes well to unseen data and maintains high performance beyond the validation set.

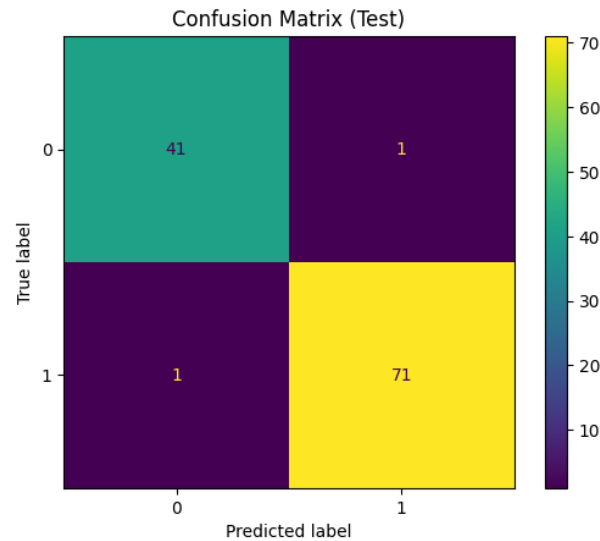


Figure 14 — Confusion matrix of the tuned Logistic Regression model evaluated on the test set.

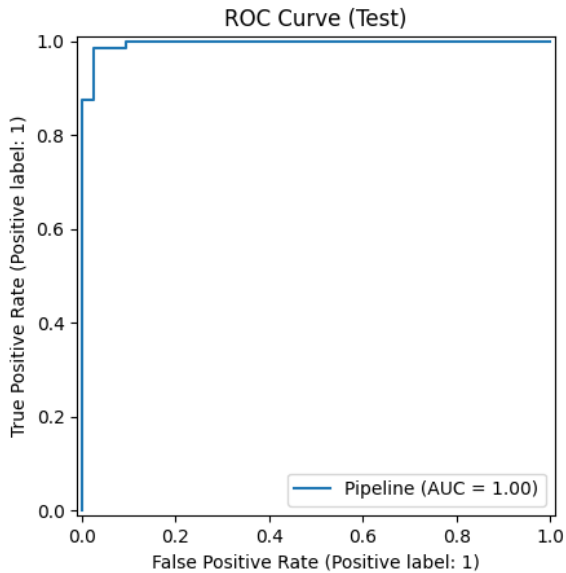


Figure 15 — ROC curve of the tuned Logistic Regression model evaluated on the test set.

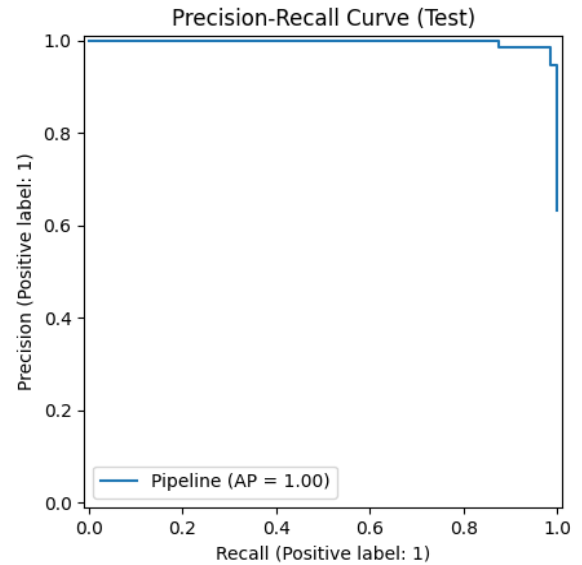


Figure 16 — Precision-Recall curve of the tuned Logistic Regression model evaluated on the test set.

The test performance is consistent with the validation results, indicating that the model does not suffer from overfitting. The small number of misclassifications confirms that the dataset is highly separable and well-suited for Logistic Regression.

Overall, this experiment demonstrates that a simple and interpretable model, when properly tuned and evaluated, can achieve very high performance on this classification task.

Step 7 — Feature Importance (Logistic Coefficients)

In this step, the importance of input features is analyzed using the coefficients of the Logistic Regression model.

Since Logistic Regression is a linear model, each feature is associated with a coefficient that reflects its contribution to the

prediction.

Features with larger absolute coefficient values have a stronger influence on the model’s decision.
The top 10 most influential features are identified based on the absolute values of their coefficients.

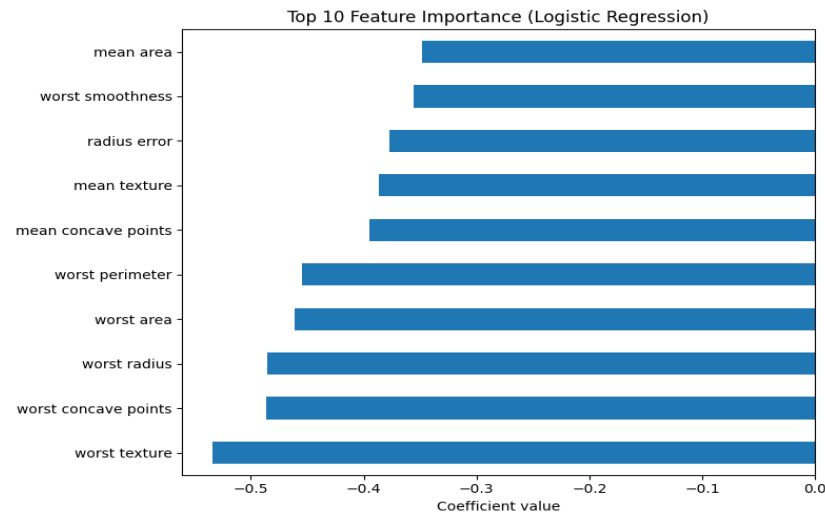


Figure 17 — Top 10 most important features based on Logistic Regression coefficients.

The feature importance analysis shows that the most influential features are primarily related to geometric and structural properties of the cell nuclei, such as texture, radius, area, and concavity.

In particular, features like "worst texture", "worst concave points", and "worst radius" have the highest absolute coefficients, indicating that they play a major role in distinguishing between malignant and benign tumors.

All top features have negative coefficients, which indicates that higher values of these features are associated with the malignant class. This is consistent with medical expectations, as malignant tumors typically exhibit more irregular shapes, higher texture variation, and greater structural complexity.

These findings confirm that the model relies on meaningful and interpretable features, contributing to both its high performance and its transparency.

Step 8 — Conclusion

In this project, an end-to-end machine learning pipeline was developed for the classification of breast cancer tumors using the UCI Wisconsin Diagnostic dataset.

The analysis started with exploratory data analysis, which revealed that the dataset is well-structured, moderately imbalanced, and partially linearly separable. These observations motivated the use of Logistic Regression as a baseline model.

A stratified train/validation/test split was applied to ensure fair evaluation and prevent data leakage. A baseline Logistic Regression model with feature scaling achieved very high performance, with accuracy close to 99% on the validation set.

Hyperparameter tuning using GridSearchCV showed that the baseline model was already close to optimal, with only minor improvements. Learning and validation curves confirmed that the model has low bias and low variance, indicating good generalization.

The final evaluation on the test set demonstrated strong and consistent performance, confirming that the model generalizes well to unseen data.

Finally, feature importance analysis highlighted that a small number of meaningful features, related to shape and texture characteristics of cell nuclei, are sufficient to achieve high classification accuracy.

Overall, this project demonstrates that a simple and interpretable model such as Logistic Regression can achieve excellent performance on this dataset, making it a strong baseline for medical classification tasks.

Future work could explore more complex models or additional techniques to further improve performance and analysis.